

Full-Stack Banking Application – DevOps Deployment & Automation

Follow the following step:

Clone the Github Repo:

```
https://github.com/krushjagdale1/FullStackBankingApplication-Krushna.git
```

Go to project folder using:

```
cd FullStackBankingApplication-Krushna/
```

```
ls -lhrt - check all file
```

Build the dockerfile using command:

Build the frontend:

```
docker build -f frontend/Dockerfile -t banking-frontend:latest .
```

Build the Backend:

```
docker build -f backend/Dockerfile -t banking-backend:latest .
```

validate it :

```
docker compose config
```

After building the Dockerfile check images :

```
docker images
```

Create a Docker Network :

To enable communication between the Frontend, Backend, and MongoDB containers, create a dedicated Docker bridge network.

```
docker network create banking-network
```

```
docker network ls
```

Run MongoDB Container :

```
docker run -d --name mongodb --network banking-network -27017:27017  
mongo:7
```

Run Backend Container :

```
docker run -d --name banking-backend --network banking-network  
-e MONGODB_URI=mongodb://mongodb:27017/banking  
-p 3001:3001  
banking-backend:latest
```

Run Frontend Container :

```
docker run -d --name banking-frontend --network banking-network -p  
80:80 banking-frontend:latest
```

Verify all containers are up and running:

```
docker ps
```

Install kube-state-metrics on the Kubernetes:

```
git clone https://github.com/kubernetes/kube-state-metrics.git  
cd kube-state-metrics  
kubectl apply -f examples/standard
```

ELK & Filebeat installation file had uploaded on Github: